

Indian Institute of Technology Madras

DA5400W - Foundations of machine learning

Tutorial and Assignment I

Due date: February 2, 2026, 10 pm

Instruction

1. Assignment shall be submitted on the due date. Late submissions will not be entertained. Any submission through an email will have a -20 mark penalty.
2. All the assignments must be the student's own work.

Problems

1. The following probability distribution function can describe the growth rate of finance:

$$f(x) = \frac{1}{\alpha^2} x e^{-x/\alpha} \quad (1)$$

with $\alpha \in (0, \infty)$ and $x \in [0, \infty)$. Find an estimate of the parameter α using the maximum likelihood estimators and the method of moments given the datasets $x_1, x_2, \dots, x_n$.

2. Consider a process for which a random sample $X_1, X_2, \dots, X_9$ is collected to understand the population properties. The population mean and variance are μ and σ^2 . Mr and Mrs. Stat have proposed two estimators as follows:

$$\begin{aligned} \theta_{Mr} &= \frac{X_1 + X_3 + 4X_5}{6} \\ \theta_{Mrs} &= \frac{X_2 - X_6 + 2X_7 + 3X_4}{5} \end{aligned} \quad (2)$$

Which is the best estimator and why?

3. The probability distribution functions of the Weibull distribution and the Rayleigh distribution are given below:

$$\text{Weibull distribution: } f(x; k, \lambda) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, x \geq 0$$

$$\text{Rayleigh distribution: } f(x; \sigma) = \frac{x}{\sigma^2} e^{-x^2/(2\sigma^2)}, x \geq 0$$

Use the dataset (link) provided to estimate the parameter λ in Weibull distribution using maximum likelihood estimation (MLE) (assume $k = 2$). Use the property of invariance of MLE to estimate the parameter σ of Rayleigh distribution.

4. Find the maximum likelihood estimate of the parameter θ of the following probability distribution function:

$$f(y; \theta) = \frac{3y^2}{\theta^3} \quad (3)$$

with $\theta \in (0, \infty)$ and $y \in [0, \theta]$ using the data $y_1, y_2, \dots, y_n$.

5. Dr. AAA collects samples of cancer patients to estimate the mean expression levels of an oncogene. Due to technical limitations, (s)he can collect only 20 samples per day and measure the expression levels of the oncogene. It has been known that the gene expression levels follow normal distribution with standard deviation 8 ($\sim \mathcal{N}(\mu, \sigma = 8)$). Help him/her in estimating the mean gene-expression value using recursive Bayesian estimation. The dataset (link) provided has gene expression levels of the oncogene collected for 10 days. Do the following:

- i) Assume the prior distribution of μ to be a normal distribution. You can take the sample mean of Day 1 samples and variance as prior parameters
- ii) Estimate the posterior distribution of μ using samples from Day 1
- iii) Update the priors and repeat step ii) using data from each of the days
- iv) Plot the probability distribution of the mean of gene expression level each time after the update

6. Consider a random variable C with mean μ and standard deviation σ . Two experiments are performed to collect two random samples with n_1 and n_2 sample sizes. The sample means for these experiments are $\bar{C}_1$ and $\bar{C}_2$. Then, show that an estimator for the sample mean

$$\bar{C} = a\bar{C}_1 + (1 - a)\bar{C}_2$$

is an unbiased estimator for the mean μ .

7. Find the local extrema of the following functions and classify the points as minimum or maximum.

- (a) $f(x) = 4x^3 - 3x^2 + 2x - 1$
- (b) $f(x) = \sin x + \cos x$
- (c) $f(x) = \frac{x^2-1}{x}$

8. Show that the function $f(x_1, x_2) = 8x_1 + 12x_2 + x_1^2 - 2x_2^2$ has only one stationary point and that it is neither a maximum nor a minimum, but a saddle point. Sketch the contour line of f .

9. Determine the stationary points and classify their nature for the function, $f(x) = x^4 + y^4 - 36xy$

10. Solve the following optimization problems by hand(s) and also draw the feasible regions:

- Find the maximum of the following function:

$$f(x) = 1 - 8x + 2x^2 - \frac{10}{3}x^3 + \frac{1}{4}x^4 + \frac{4}{5}x^5 - \frac{1}{6}x^6$$

$$f(x_1, x_2) = x_1 + x_2 \text{ subject to } x_1^2 + x_2^2 - 1 = 0$$

- Verify the KKT conditions and find the Lagrange multipliers for the following function at $x = (1, 0)$

$$\left(x_1 - \frac{3}{2}\right)^2 + \left(x_2 - \frac{1}{8}\right)^4$$

subject to

$$\begin{bmatrix} 1 - x_1 - x_2 \\ 1 - x_1 + x_2 \\ 1 + x_1 - x_2 \\ 1 + x_1 + x_2 \end{bmatrix} \geq 0$$

- Find the minimum of the following function:

$$f(x) = (x_1 - 1)^2 + x_2^2$$

subject to

$$x_1 - x_2^2 \leq 0$$